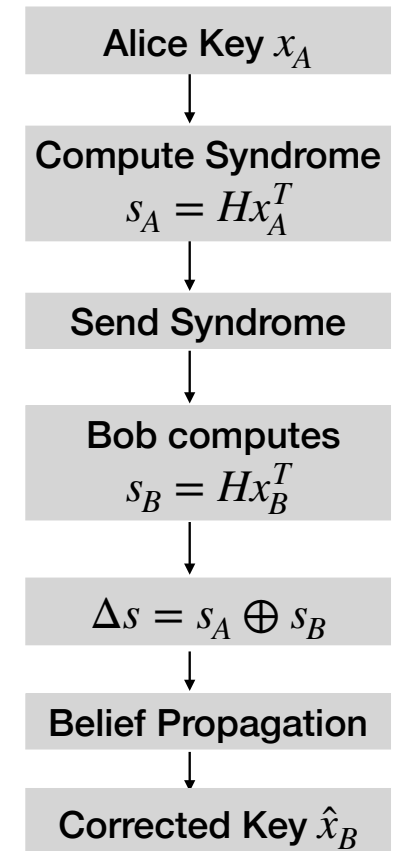
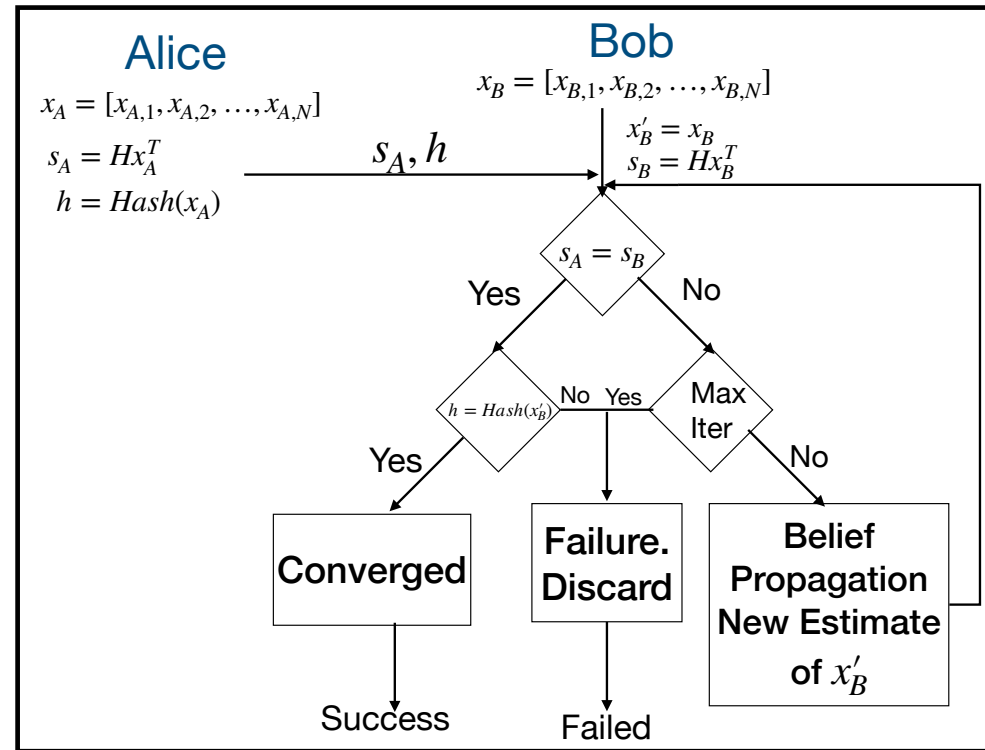




Low-Density Parity Check Codes

- **Alice computes:** $s_A = Hx_A^T$ (and sends it to Bob) where H : sparse parity-check matrix and x_A : Alice's sifted key
- **Bob computes:** $s_B = Hx_B^T$ and the syndrome difference $\Delta s = s_A \oplus s_B$, which identifies parity inconsistencies caused by errors.
- Let $x_A = x_B + e$, then $\Delta s = s_A \oplus s_B = He^T$
- Bob solves for e using $\Delta s = He^T$ and x_B using Belief Propagation Algorithm
- Bob corrects the errors in x_B using the estimated e



LDPC Information Reconciliation



1. Reconciliation Efficiency:

$$f_{EC} = \frac{\text{Actual Information disclosed}}{\text{Shannon Limit}}$$

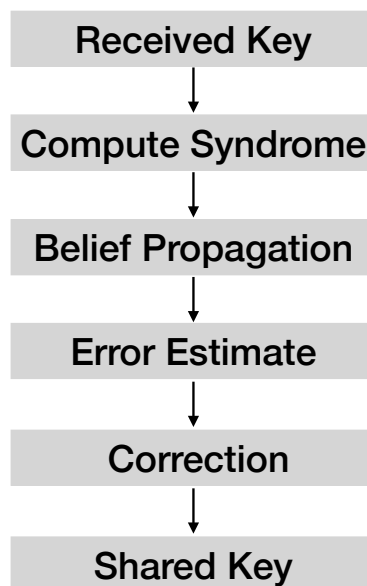
$$f_{EC} = \frac{leak_{EC}}{nh(Q)}$$

where:

n = sifted key length, Q = QBER

$h(Q)$ = binary entropy function

$leak_{EC}$ = actual information leaked during reconciliation.



• Advantages:

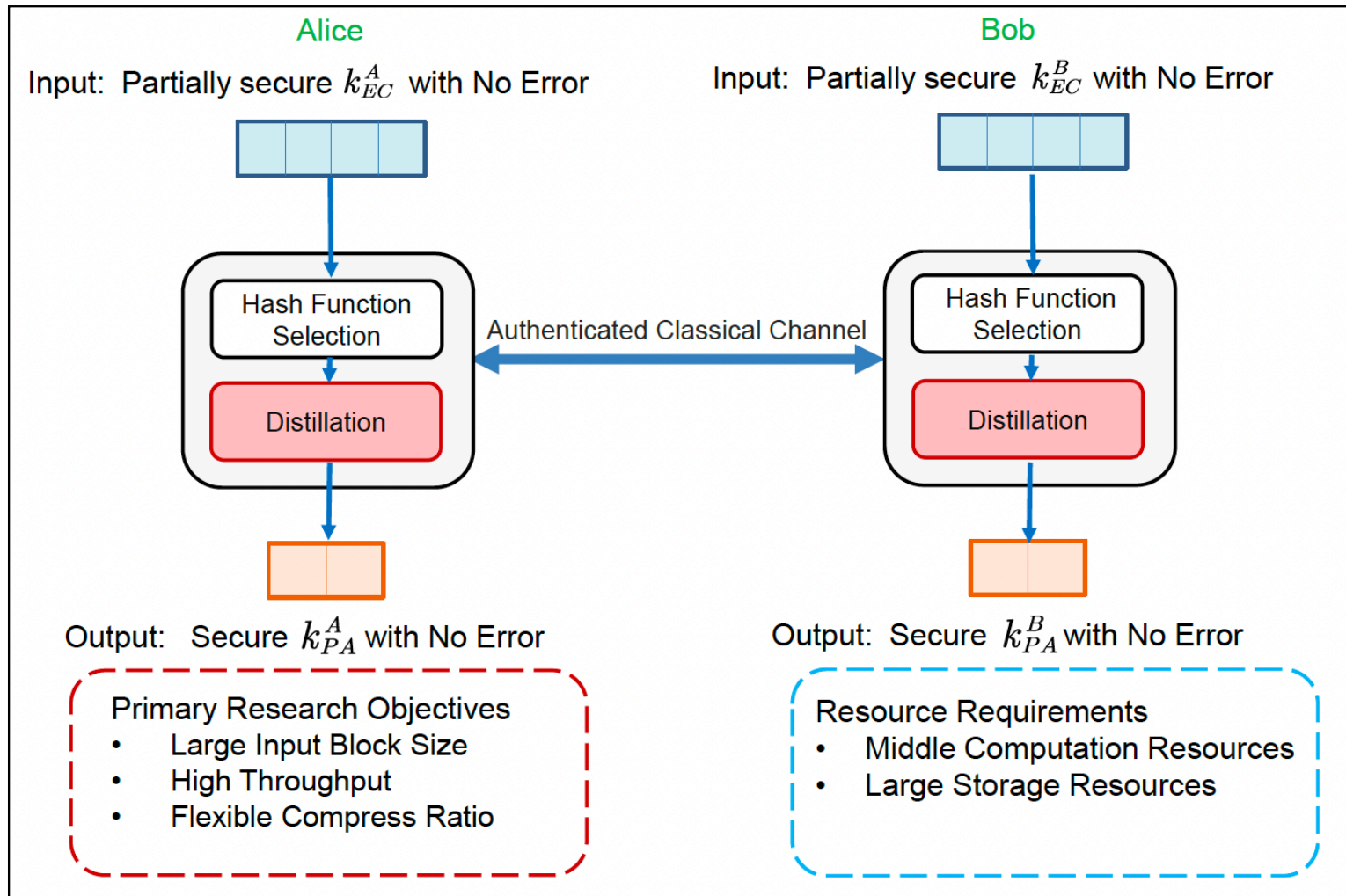
- ✓ Near Shannon Limit
- ✓ Very Low Leakage
- ✓ One-way Communication
- ✓ High Throughput
- ✓ Long Distance Compatible

2. For $H \in \mathbb{F}_2^{m \times n}$: $leak_{EC} \approx m$

3. If LDPC Code rate: $R = \frac{k}{n}$: then
 $leak_{EC} \approx m = n - k = n(1 - R)$



Privacy Amplification





Privacy Amplification

- Suppose Alice and Bob share:

1010110011010101

- Eve may know some bits due to information leakage during error correction step
- Instead of using the entire key, Alice and Bob compress it into a shorter key:

1010110011010101

↓

Hash

↓

110101

- The new key becomes **effectively unknown** to Eve.



Privacy Amplification

- **Security Objective**

Let: K be the reconciled key.

E represent Eve's information.

- **Goal:** Generate a new key: K_{final}

such that

$$I(K_{final}; E) \approx 0$$

where $I(\cdot)$ denotes mutual information.

- **Role of Privacy Amplification**

- Privacy amplification compresses the reconciled key K into a shorter key K_{final} :

$$K_{final} = TK$$

where T is a hash matrix (Toeplitz, modified Toeplitz, universal hash family, etc.).

- The amount of compression determines the final key length

- **Concept of Compression**

- Privacy amplification intentionally reduces key length.

Long Key



Universal Hash Function



Short Secure Key

- **Tradeoff:** Length ↓
Security ↑



Privacy Amplification

- **Universal Hashing:** A hash family is universal if:

$$P[h(x) = h(y)] \leq \frac{1}{2^m}$$

For: $x \neq y$ where m = output length.

- **Matrix-Based Hashing:** This ensures that Eve cannot predict the final key.

- Toeplitz Hash Matrix: A Toeplitz matrix has constant diagonals (requires less no. of bits to represent)

$$T = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ a_5 & a_1 & a_2 & a_3 \\ a_6 & a_5 & a_1 & a_2 \\ a_7 & a_6 & a_5 & a_1 \end{bmatrix}$$

- Final Key: $K_{\text{final}} = TK$ (multiplication can be accelerated using FFT)

- **Reconciled key:**

1011010110011010

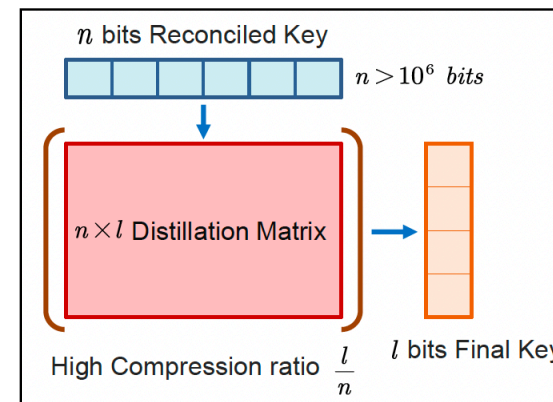
- **Hash Function:**

Universal Hash

- **Output:**

11001101

- Key length reduced
- Eve's information drastically reduced





Authentication

- Authentication ensures the integrity and security of the communication between Alice and Bob.
- Classical Channel is authenticated using some preshared keys
- A part of final QKD keys (from previous rounds) is consumed for authentication

